

Data Science - Core Concepts

Data Science is an interdisciplinary field that combines statistics, mathematics, programming, and domain knowledge to extract meaningful insights from data. Data scientists work with both structured data (tables, databases) and unstructured data (text, images, audio, video).

The Data Science Workflow

1. Data Collection - Gathering raw data from various sources.
2. Data Cleaning - Handling missing values, duplicates, and outliers.
3. Exploratory Data Analysis (EDA) - Understanding patterns using statistics and plots.
4. Feature Engineering - Transforming raw data into useful features for models.
5. Model Building - Training machine learning models on prepared data.
6. Model Evaluation - Measuring accuracy, precision, recall, and F1-score.
7. Deployment - Serving the model in a real-world application.

Key Python Libraries for Data Science

- NumPy : Numerical computing with arrays and matrices.
- pandas : Data manipulation and analysis with DataFrames.
- Matplotlib and Seaborn : Data visualization.
- scikit-learn : Machine learning algorithms and utilities.
- TensorFlow and PyTorch : Deep learning frameworks.

Statistics in Data Science

Statistics is the foundation of data science. Key concepts include:

- Mean, Median, Mode : Measures of central tendency.
- Variance and Standard Deviation : Measures of spread.
- Probability Distributions : Normal, Binomial, Poisson.
- Hypothesis Testing : t-test, chi-square test, ANOVA.
- Correlation and Regression : Measuring relationships between variables.

Big Data Technologies

When data is too large to fit in memory, we use distributed systems such as:

- Apache Hadoop : Distributed storage and processing.
- Apache Spark : Fast in-memory distributed computing.
- Apache Kafka : Real-time data streaming.